Clovis SFEIR

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French (Native) – English (C1)

Profile

IMT Atlantique engineering student specialising in AI/ML and performance-critical systems for industrial infrastructure. Built five systems (Python/Rust/C++20) benchmarked against industry references or labeled ground-truth: a from-scratch BM25 retrieval engine for engineering documents (**MRR 0.944**), an IIoT predictive-maintenance pipeline (**0/10 false positives**), an Ornstein-Uhlenbeck crude-spread engine, and an IEC-104 ICS intrusion detector with **100% attack recall**.

Education

IMT Atlantique – Engineering Degree in Software Engineering (Apprenticeship) *Sep 2025 – Jun 2028*
Nantes, France *Algorithms & Data Structures, OS, Computer Networks, Distributed Systems, Cloud Computing, AI*

Bachelor in Computer Science *Sep 2022 – Jun 2025*
La Rochelle, France ***Class rank: 1/85*** *| Software Engineering, Databases, Cybersecurity*

HRT Macro Placement Challenge 2026 (Hudson River Trading/Partcl) — **24th/124** (top 20%); hybrid analytical placer: force-directed global placement with timing-aware legalization; proxy cost **1.095** beats SA baseline (2.125) and RePlAce (1.458); field includes Stanford, Princeton, ETH Zürich.

Projects

Engineering Docs RAG *Python, BM25, information retrieval*

- From-scratch Okapi BM25 retrieval engine (chunking + ranking) for engineering documents (datasheets, HAZOP-style procedures); pluggable extractive/LLM generator interface, zero paid API calls in tests/CI/demo by design.
- Hand-derived BM25 scoring verified to 9 significant figures; measured **precision@1 0.90**, recall@5 1.00, **MRR 0.944** on a 40-question labeled set; ~3,330 docs/s ingest; 53 tests, zero runtime dependencies.

IIoT Predictive Maintenance *Rust, tokio, EWMA/CUSUM*

- Async telemetry pipeline (`tokio` mpsc) fuses EWMA + CUSUM control charts across 3 sensor channels to detect developing bearing-wear signatures, with linear-trend remaining-useful-life estimation.
- Sensor-fusion tuning (2-of-3 channels + widened control limits) cut false positives from a 0.41–0.50 precision baseline to **0/10 across 10 seeds**; precision **1.0000**, recall 0.9642, F1 0.9818, 43-sample latency; **1.6 M samples/s**; 26 tests, 0 failures.

Crude Oil Spread Trading Engine *C++20, stochastic processes, order book*

- Models refining margins (3:2:1 crack spread, generalized N:M:K) and mean-reverting inter-crude spreads via an Ornstein-Uhlenbeck process (Euler-Maruyama) with a rolling z-score signal, matched by a price-time-priority order book.
- OU stationary distribution verified against theory (500K-step sim); crack-spread formula checked to 10^{-9}; order-book submit at **720 ns** (non-crossing)/**1.24 µs** (crossing); 58 assertions/21 tests, ASan+UBSan clean.

Reservoir Diffusivity PDE Solver *C++20, OpenMP, finite-difference*

- Explicit FTCS and implicit Crank-Nicolson finite-difference schemes for the reservoir diffusivity (pressure-transient) equation; verified via grid-refinement convergence study against the closed-form erfc analytical solution – error ratio converges to **4.0** as dx halves (textbook 2nd-order accuracy) across 5 refinements.
- OpenMP-parallelized stencil/assembly loops; measured (not assumed) scaling identifies the Amdahl's-law-bounded sequential Thomas solve and memory-bandwidth-bound behavior below L3 size; 13 test cases, 40,300 assertions; Docker image runs the full suite at build time.

ICS Intrusion Detection (IEC 60870-5-104) *C++20, protocol parsing, explainable rules*

- Rule-based detector for IEC 60870-5-104 (no auth/encryption by protocol design) – the actual telecontrol protocol behind the 2016 Kyiv blackout – mapped to real attack patterns: control commands to unexpected breaker objects (the exact method used), sequence-number gaps (replay/desync), u-format/command floods.
- Parses APCI+ASDU at **39.5 ns** (**25.3 M msg/s**); end-to-end parse+detect 337 ns; **100% recall, 0 false positives** on a labeled synthetic trace (100 records, 15 attacks across 5 rules); 153 assertions/37 tests, ASan+UBSan clean.

Work Experience

Infotel *Nantes, France*
AI Engineer Apprentice – Lab IA *Sep 2025 – Jun 2028*

- Built a Rust-based autonomous software-delivery harness that takes a complete specification through architecture synthesis (subsystems, interface contracts), epic/slice decomposition, and TDD-first generation to a tested, deployable application end-to-end; automated compile+test quality gates with iterative self-repair, independent code-review pass, and cross-slice conflict detection before delivery; 928 Rust tests; validated end-to-end producing a fully working, 100%-test-passing application directly from a single specification.

Excelia Group *La Rochelle, France*
Software Developer Intern / AI Software Engineer Intern *Two internships: Jan–Mar 2025 & May–Jul 2024*

- RAG pipeline (chunking, OpenAI embeddings, flat FAISS, cross-encoder re-ranking): **40% reduction** in lookup time vs. keyword-search; cut p95 latency 30–40% via embedding cache + asyncio parallelisation; Prometheus/Grafana, Docker, GCP.
- Real-time interview simulator (Next.js/React, GPT-4o, ElevenLabs TTS): streaming responses, scenario-branching state machine, A/B prompt-scoring harness.

Technical Skills

Languages	C++20, Python, Rust, TypeScript, SQL
Perf / Systems	OpenMP, cache-aware layout, acquire/release atomics, ICS/SCADA protocol parsing (Modbus, IEC-104)
ML / Quant	BM25/information retrieval, EWMA/CUSUM change-detection, Kalman filter, Ornstein-Uhlenbeck, finite-difference PDE, LightGBM/XGB/CatBoost, FAISS
Infrastructure	Docker, tokio async, Prometheus/Grafana, GitHub Actions CI, GCP, Redis, ZeroMQ
Frameworks	Flask, Node.js/Express, React/Next.js, REST/gRPC